



Increasing the Flyback Converter Input Voltage to 48V

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Project Summary

The primary project goal is to increase the input voltage of the flyback converter from 18 volts to 48 volts while maintaining the 10 volt output.

Preparing the lab hardware for the increased voltage input required several modifications:

- Building a new transformer for the required system parameters
- Creating an auxiliary winding & associated power circuitry to supply the UC2844 with an acceptable operating voltage.
- Designing an appropriate snubber and clamp for the higher peak voltage
- Ensuring all other components' ratings are within the higher voltage limit

Overall, we successfully realized a 48V-to-10V flyback converter that operated at a higher 81.6% efficiency compared to the 18V-to-10V equivalent.

Math & System Parameters

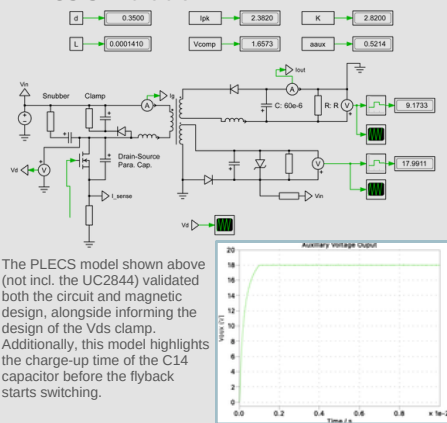
The new transformer's design followed the same parameters as previous magnetics designed in class. Namely,

- A critical inductance ratio of $\alpha = 0.7$
- A duty cycle of $D = 35\%$.

These specifications and our input/output voltage determined the turns ratio, 0.29. This turns ratio informed computation of the representative output voltage and resistance, which in turn calculated a critical inductance of $201\mu\text{H}$, and a subsequent designed magnetizing inductance $141\mu\text{H}$. This represents a significantly higher inductance compared to the lower input voltage. Why? Given the current rises at a rate proportional to voltage over inductance, the higher inductance is needed to maintain a lower current to achieve the same power at a higher voltage.

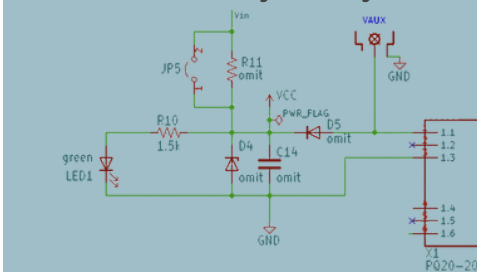
Simulations in PLECS showed that the peak drain voltage approaching 190V, which presented a serious possibility of destroying the 200V-rated MOSFET used. As such, a new clamp utilizes a lower capacitance and higher resistance ($33\text{nF} \ \& \ 3.9\text{k}\Omega$). The lower capacitance allows the voltage across the clamp to rise quicker & reach the higher clamping voltage, while the larger resistor limits the average power dissipated.

PLECS Simulation



The PLECS model shown above (not incl. the UC2844) validated both the circuit and magnetic design, alongside informing the design of the Vds clamp. Additionally, this model highlights the charge-up time of the C14 capacitor before the flyback starts switching.

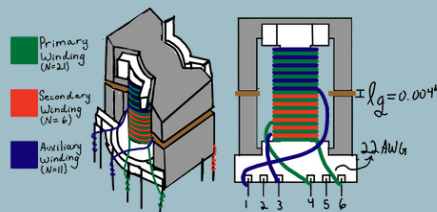
Transformer & Auxiliary Circuitry



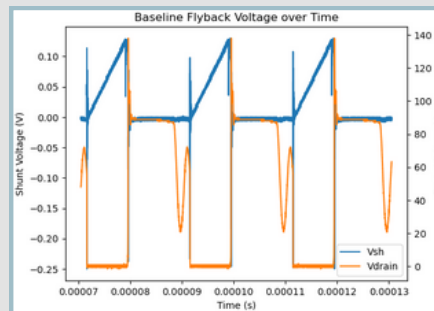
In order to realize the auxiliary circuit, jumper JP5 was removed and a 1500Ω resistor was added to R11. A 18V Zener diode and a $22\mu\text{F}$ capacitor were added to D4 and C14. R10 was removed to reduce power losses. To power the circuit, the auxiliary winding of the transformer was connected to pins 1 & 3 of the bobbin.

The transistor was made by first turning the primary and auxiliary with a bifilar winding, then continuing on with the primary and secondary still with a bifilar winding to minimize leakage inductance.

All windings were done clockwise with the primary going from 6 to 4, the secondary from 10 to 7 and the auxiliary going from 1 to 3. The turn numbers for the primary, secondary and auxiliary are 21, 6, and 11 respectively and 22 AWG wire was used for each.



Real-World Prototype



After the auxiliary circuit was designed, the transformer constructed, and all components evaluated, the board was re-soldered and populated with its new components.

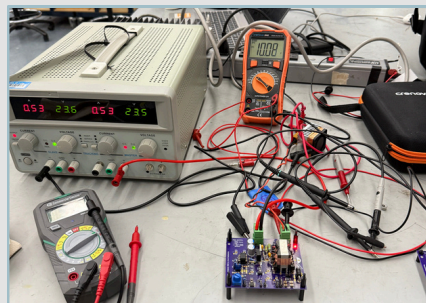
To achieve an input voltage of 48V, the power supply was set to parallel operation and 24V output was set to both sides of the supply.

Given the higher voltage, extra safety precautions were taken. Safety glasses were worn at all times. The person interacting with the board sat on a chair for insulation. The potentiometer was turned with a screw driver, and both probes of the multimeter were held in one hand when checking component to reduce damage of a potential shock.

Initially, there were issues with providing enough current to the UC2844 to power the timing circuit. After removing LED1 and reducing the value of R11, the auxiliary circuit got enough power to start up the switching.

The peak current was 2.54A which slightly higher than the simulation estimate of 2.38A . The drain voltage peaked at around 140V , this was much higher than the max intended voltage of 100V , but it is still well below the MOSFET limit of 200V .

During operation, our circuit had an input power from the supply of 24.91W with an output power of 20.32 . This gives the converter an efficiency of 81.6% . This is slightly larger than the circuit operating at 18V input voltage which had an efficiency of 78.1% .



Other Component Considerations

- The smaller ceramic input filtering capacitors are only rated for 35V . As such, they were removed in favor of the larger 100V electrolytic capacitors.
- Adjusting the clamp voltage ensured that the 200V MOSFET drain-to-source voltage rating was respected.
- Increasing the resistor on LED1 prevented start-up current from being drawn away from the UC2844. The LED was taken out to allow a smaller auxiliary power draw from the input rail.

Further Improvements

The first major aspect to improve is the voltage limit of the auxiliary circuit. During testing, we eventually blew out our Zener diode and at that point, we were unable to properly regulate our UC2844 supply voltage.

This issue could be fixed in two ways. The first would be to use a higher rated or slightly higher voltage diode to prevent it from blowing again. The second would be to increase R11 or decrease the auxiliary inductance to limit the additional current coming into the circuit that needs to be discharged through the Zener.

Another improvement would be to reduce the resistance of the Vcomp potentiometer or include a type II compensator. The higher input voltage has a lower peak current which makes adjusting the voltage less precise and the two options presented above would correct this.

Finally, the clamp of the circuit was not functioning as well as intended and redesigning the clamp would ensure better protection of the circuit from spikes in voltage.

Conclusion

Overall, the project was a success as we were able to reach our desired goal of a 10V output with a 48V input.

With this achievement, came many lessons learned, short falls overcome, and steps to consider in the future.

Through this project, we gained tremendous knowledge in further designing power converters under much more difficult requirements and the design and implementation of the auxiliary circuit was an incredible opportunity for this learning to take place.

After the initial construction, there were many issues discovered in the circuit. The primary fault of the circuit was a lack of power initially delivered to the UC2844, which meant it wasn't able to begin switching the MOSFET to start up the transformer's power transfer.

In the future, certain parts should be better sourced to handle their power loads. We saw issues with this in the Zener diode and in R11 which saw a higher power than its rate 0.25W .

We also had some key takeaways from this project. One key lesson is that there was a slight improvement in the efficiency of the circuit. This could come from the removal of the LED and reduced conduction losses on the primary side of the circuit caused by a lower peak current at higher input voltages. Finally, when designing at high power it is crucial to carefully inspect all elements of the circuit to ensure that can operate as intended.

This project was a beneficial challenge as it forced us to extend our knowledge gained in Power Electronics to solve novel problems and troubleshoot & investigate issues to redesign and ultimately achieve functionality.